

Supplementary Material for *Bifurcation of Limit Cycles in the Coupling of Two van der Pol Type Oscillators*

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Description

This document contains the Mathematica commands used in the symbolic computations appearing in the proofs of Theorems 2 and 3 of the paper *Bifurcation of Limit Cycles in the Coupling of Two van der Pol Type Oscillators*. A complete Mathematica notebook containing all inputs and outputs is also available in the GitHub repository:

<https://github.com/luizfgoncalves11/research>

Remark. The commands below are intended to document the symbolic computations used in the paper. They are not optimized for computational efficiency, but rather for transparency and reproducibility. The symbolic computations serve as an aid for performing lengthy algebraic manipulations. The proofs do not rely on computer-assisted verification, and all mathematical arguments can be checked by hand from the formulas presented in the manuscript.

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1 Computations for Theorem 2

We here provide the Mathematica commands used in the symbolic computations leading to the proof of Theorem 2.

```
(* ===== *)  
(* Supplementary computations for Theorem 2 *)  
(* ===== *)  
  
Quit[]  
  
(* Perturbation functions *)
```

```

f[x_,y_,z_,w_] := y(x^2-r1)-a(z-x);
g[x_,y_,z_,w_] := w(z^2-r2)-b(x-z);

(* Perturbed coupled Van der Pol system considered in Theorem 2 *)

systempert = {
x'[t] -> y[t],
y'[t] -> -x[t] + \[Epsilon]*f[x[t],y[t],z[t],w[t]],
z'[t] -> w[t],
w'[t] -> -z[t] + \[Epsilon]*g[x[t],y[t],z[t],w[t]]
};

(* Change of variables defined by equations (9) and (10),
based on the first integrals introduced in equation (8) *)

mudanca = {
x[t] -> Sqrt[2 R] Cos\[Theta],
y[t] -> Sqrt[2 R] Sin\[Theta],
z[t] -> (T Cos\[Theta] - S Sin\[Theta])/(Sqrt[2] Sqrt[R]),
w[t] -> (S Cos\[Theta] + T Sin\[Theta])/(Sqrt[2] Sqrt[R]),
R[t] -> R
};

(* Computing the system in (R,[Theta],S,T) *)

Rp =
x[t] x'[t] +
y[t] y'[t] /.
systempert /. mudanca //
Simplify // FullSimplify;

Sp =
x[t] w'[t] +
w[t] x'[t] -
z[t] y'[t] -
y[t] z'[t] /.
systempert /. mudanca //
FullSimplify;

Tp =
y[t] w'[t] +
z[t] x'[t] +
w[t] y'[t] +
x[t] z'[t] /.
systempert /. mudanca //
FullSimplify // Factor;

thetaP =
Collect[
(x[t] y'[t] -
y[t] x'[t])/(2 R[t]) /.
systempert /. mudanca //
FullSimplify,
\[Epsilon]

```

```

];

(* Taking \[Theta] as independent variable and extracting the
0(\[Epsilon]) terms *)

dRdTheta = Rp/thetaP;
dSdTheta = Sp/thetaP;
dTdTheta = Tp/thetaP;

seriesdRdTheta =
Normal[Series[dRdTheta, {\[Epsilon],0,2}]]
// FullSimplify;

G11 =
Coefficient[
Collect[seriesdRdTheta,\[Epsilon]],
\[Epsilon]
] // Simplify;

seriesdSdTheta =
Normal[Series[dSdTheta,{\[Epsilon],0,2}]]
// FullSimplify;

G22 =
Coefficient[
Collect[seriesdSdTheta,\[Epsilon]],
\[Epsilon]
] // FullSimplify;

seriesdTdTheta =
Normal[Series[dTdTheta,{\[Epsilon],0,2}]]
// FullSimplify;

G33 =
Coefficient[
Collect[seriesdTdTheta,\[Epsilon]],
\[Epsilon]
] // FullSimplify;

check = {
FullSimplify[G11 + Rp/\[Epsilon]],
FullSimplify[G22 + Sp/\[Epsilon]],
FullSimplify[G33 + Tp/\[Epsilon]]
};

(* First-order averaged system *)

{G11,G22,G33};

(* Computing the averaged functions *)

f1 =
(1/(2 Pi))
Integrate[G11,{\[Theta],0,2 Pi}]

```

```

// FullSimplify // Factor;

f2 =
(1/(2 Pi))
Integrate[G22,{\[Theta],0,2 Pi}]
// FullSimplify;

f3 =
(1/(2 Pi))
Integrate[G33,{\[Theta],0,2 Pi}]
// FullSimplify;

ff = Numerator[{f1,f2,f3}];

(* Multiplying the averaged functions by -1
does not alter their common zeros *)

(* Solving the first averaged equation for S *)

solS =
Solve[ff[[1]]==0,S][[1]]
// FullSimplify;

(* Substituting S into the remaining averaged equations *)

eq2 =
Numerator[
Together[(ff[[2]] /. solS)]
] // Expand;

eq3 =
Numerator[
Together[(ff[[3]] /. solS)]
] // Expand;

Collect[
FullSimplify[{eq2,eq3}],
{T,R}
];

(* Computing the resultant *)

resT =
Resultant[eq2,eq3,T]
// Factor;

P =
Factor[
resT/(256 a^10 R^4)
];

(* Degree of the polynomial P(R) *)

Exponent[P,R]

```

2 Computations for Theorem 3

We here provide the Mathematica commands used in the symbolic computations leading to the proof of Theorem 3.

```
(* ===== *)
(* Supplementary computations for Theorem 3 *)
(* ===== *)

Quit[]

(* Perturbation functions *)

f[x_,y_,z_,w_] :=
Sin[x^2+y^2] (a2*x+b2*y+c2*z+d2*w);

g[x_,y_,z_,w_] :=
Sin[z^2+w^2] (a4*x+b4*y+c4*z+d4*w);

(* Perturbed coupled Van der Pol system considered in Theorem 3 *)

systempert = {
  x'[t] -> y[t],
  y'[t] -> -x[t] + \[Epsilon]*f[x[t],y[t],z[t],w[t]],
  z'[t] -> w[t],
  w'[t] -> -z[t] + \[Epsilon]*g[x[t],y[t],z[t],w[t]]
};

(* Change of variables defined by equations (9) and (10),
based on the first integrals introduced in equation (8) *)

mudanca = {
  x[t] -> Sqrt[2 R] Cos\[Theta],
  y[t] -> Sqrt[2 R] Sin\[Theta],
  z[t] -> (T Cos\[Theta] - S Sin\[Theta])/
(Sqrt[2] Sqrt[R]),
  w[t] -> (S Cos\[Theta] + T Sin\[Theta])/
(Sqrt[2] Sqrt[R]),
  R[t] -> R
};

(* Computing the system in (R,\[Theta],S,T) *)

Rp =
x[t] x'[t] +
y[t] y'[t] /.
systempert /. mudanca //
Simplify // FullSimplify;

Sp =
x[t] w'[t] +
w[t] x'[t] -
z[t] y'[t] -
```

```

y[t] z'[t] /.
systempert /. mudanca //
FullSimplify;

Tp =
y[t] w'[t] +
z[t] x'[t] +
w[t] y'[t] +
x[t] z'[t] /.
systempert /. mudanca //
FullSimplify // Factor;

thetaP =
Collect[
(x[t] y'[t] -
y[t] x'[t])/(2 R[t]) /.
systempert /. mudanca //
FullSimplify,
\[Epsilon]
];

(* Taking \[Theta] as independent variable and extracting the
O(\[Epsilon]) terms *)

dRdTheta = Rp/thetaP;
dSdTheta = Sp/thetaP;
dTdTheta = Tp/thetaP;

seriesdRdTheta =
Normal[Series[dRdTheta,{\[Epsilon],0,2}]]
// FullSimplify;

G11 =
Coefficient[
Collect[seriesdRdTheta,\[Epsilon]],
\[Epsilon]
] // Simplify;

seriesdSdTheta =
Normal[Series[dSdTheta,{\[Epsilon],0,2}]]
// FullSimplify;

G22 =
Coefficient[
Collect[seriesdSdTheta,\[Epsilon]],
\[Epsilon]
] // FullSimplify;

seriesdTdTheta =
Normal[Series[dTdTheta,{\[Epsilon],0,2}]]
// FullSimplify;

G33 =
Coefficient[

```

```

Collect[seriesdTdTheta,\[Epsilon]],
\[Epsilon]
] // FullSimplify;

check = {
    FullSimplify[G11 + Rp/\[Epsilon]],
    FullSimplify[G22 + Sp/\[Epsilon]],
    FullSimplify[G33 + Tp/\[Epsilon]]
};

(* First-order averaged system *)

{G11,G22,G33};

(* Computing the averaged functions *)

w1 =
(1/(2 Pi))
Integrate[G11,{\[Theta],0,2 Pi}]
// FullSimplify // Factor;

w2 =
(1/(2 Pi))
Integrate[G22,{\[Theta],0,2 Pi}]
// FullSimplify;

w3 =
(1/(2 Pi))
Integrate[G33,{\[Theta],0,2 Pi}]
// FullSimplify;

ff = Numerator[{w1,w2,w3}];

(* Evaluating the averaged functions at  $R_k = k \text{ Pi}/2$  *)

wk =
FullSimplify[
ff /. R -> k Pi/2,
Assumptions -> k \[Element] Integers && k > 0
];

(* Solving the linear factors *)

L1 = a4 k Pi + d4 S + c4 T;

L2 = b4 k Pi - c4 S + d4 T;

solST =
Solve[{L1==0,L2==0},{S,T}][[1]]
// FullSimplify;

rho2 =
FullSimplify[(S^2+T^2) /. solST];

```

```

arg =
FullSimplify[rho2/(k Pi)];

(* Averaged vector field *)

W = {w1,w2,w3};

(* Jacobian matrix *)

Jac = D[W,{R,S,T}];

(* Evaluation at the singular points *)

Jack =
FullSimplify[
Jac /. R -> k Pi/2 /. solST,
Assumptions -> k \[Element] Integers && k > 0
];

(* Jacobian determinant *)

DetJac =
FullSimplify[
Factor[Det[Jack]]
];

(* Nondegeneracy condition *)

CC =
a4 (c4 d2 - c2 d4) +
b4 (c2 c4 + d2 d4) -
b2 (c4^2 + d4^2);

Assuming[
CC != 0 &&
Sin[((a4^2+b4^2) k Pi)/(c4^2+d4^2)] != 0,
FullSimplify[DetJac != 0]
]

```